

48. Let $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$. Prove that

- (a) $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|$ (**Cauchy-Schwarz inequality**);
- (b) $|\mathbf{x} \cdot \mathbf{y}| = \|\mathbf{x}\| \|\mathbf{y}\|$ if and only if $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent;
- (c) $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

Solution:

- (a) If either $\mathbf{x} = \mathbf{0}$ or $\mathbf{y} = \mathbf{0}$, then the result is obvious. So, let $\mathbf{x} \neq \mathbf{0}, \mathbf{y} \neq \mathbf{0}$. Put $\mathbf{v} = \mathbf{x} - \left(\frac{\mathbf{y} \cdot \mathbf{x}}{\|\mathbf{y}\|^2}\right) \mathbf{y}$. Then we have

$$\begin{aligned} 0 &\leq \mathbf{v} \cdot \mathbf{v} = \left(\mathbf{x} - \left(\frac{\mathbf{y} \cdot \mathbf{x}}{\|\mathbf{y}\|^2}\right) \mathbf{y}\right) \cdot \left(\mathbf{x} - \left(\frac{\mathbf{y} \cdot \mathbf{x}}{\|\mathbf{y}\|^2}\right) \mathbf{y}\right) \\ &\Rightarrow \mathbf{x} \cdot \mathbf{x} - \frac{|\mathbf{x} \cdot \mathbf{y}|^2}{\|\mathbf{y}\|^2} - \frac{|\mathbf{x} \cdot \mathbf{y}|^2}{\|\mathbf{y}\|^2} + \frac{|\mathbf{x} \cdot \mathbf{y}|^2}{\|\mathbf{y}\|^4} \|\mathbf{y}\|^2 \geq 0 \\ &\Rightarrow \|\mathbf{x}\|^2 - \frac{|\mathbf{x} \cdot \mathbf{y}|^2}{\|\mathbf{y}\|^2} \geq 0 \\ &\Rightarrow |\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|. \end{aligned}$$

- (b) We see in part (a) that $|\mathbf{x} \cdot \mathbf{y}| = \|\mathbf{x}\| \|\mathbf{y}\|$ iff $\mathbf{v} = \mathbf{0}$, that is, $\mathbf{x} = \left(\frac{\mathbf{y} \cdot \mathbf{x}}{\|\mathbf{y}\|^2}\right) \mathbf{y}$. Thus $|\mathbf{x} \cdot \mathbf{y}| = \|\mathbf{x}\| \|\mathbf{y}\|$ if and only if $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent.

- (c) We have

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\|^2 &= (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) \\ &= \mathbf{x} \cdot \mathbf{x} + \mathbf{x} \cdot \mathbf{y} + \mathbf{y} \cdot \mathbf{x} + \mathbf{y} \cdot \mathbf{y} \\ &= \|\mathbf{x}\|^2 + 2\operatorname{Re}(\mathbf{x} \cdot \mathbf{y}) + \|\mathbf{y}\|^2, \text{ as } \mathbf{x} \cdot \mathbf{y} + \mathbf{y} \cdot \mathbf{x} = \mathbf{x} \cdot \mathbf{y} + \overline{\mathbf{x} \cdot \mathbf{y}} = 2\operatorname{Re}(\mathbf{x} \cdot \mathbf{y}) \\ &\leq \|\mathbf{x}\|^2 + 2|\mathbf{x} \cdot \mathbf{y}| + \|\mathbf{y}\|^2 \\ &\leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\| \|\mathbf{y}\| + \|\mathbf{y}\|^2 \\ &= (\|\mathbf{x}\| + \|\mathbf{y}\|)^2. \end{aligned}$$

Hence $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

□

49. Let $V = \{[x, y, z, w]^t \in \mathbb{C}^4 : x = z + iw, y = z - w\}$. Show that V is a subspace of $\mathbb{C}^4$. Find a basis for each of V and $V^\perp$.

Solution: We see that

$$\begin{aligned} V &= \{[x, y, z, w]^t \in \mathbb{C}^4 : x = z + iw, y = z - w\} \\ &= \{[z + iw, z - w, z, w]^t \in \mathbb{C}^4 : z, w \in \mathbb{C}\} \\ &= \{z[1, 1, 1, 0]^t + w[i, -1, 0, 1]^t \in \mathbb{C}^4 : z, w \in \mathbb{C}\} \\ &= \operatorname{span}([1, 1, 1, 0]^t, [i, -1, 0, 1]^t). \end{aligned}$$

Hence V is a subspace of $\mathbb{C}^4$. Moreover, $\{[1, 1, 1, 0]^t, [i, -1, 0, 1]^t\}$ is a linearly independent set. Hence $\{[1, 1, 1, 0]^t, [i, -1, 0, 1]^t\}$ is a basis for V .

We see that $V = \text{row}(A)$, where $A = \begin{bmatrix} 1 & -1 & 1 & 0 \\ i & -1 & 0 & 1 \end{bmatrix}$. Therefore $V^\perp = \text{row}(A)^\perp = \text{null}(\overline{A})$. We find that $RREF(\overline{A}) = \begin{bmatrix} 1 & 0 & \frac{i+1}{2} & \frac{i+1}{2} \\ 0 & 1 & \frac{1-i}{2} & -\frac{i+1}{2} \end{bmatrix}$. From this, a basis for $V^\perp = \text{null}(\overline{A})$ is obtained as $\left\{ \begin{bmatrix} -\frac{i+1}{2} & \frac{i+1}{2} & 0 & 1 \end{bmatrix}^t, \begin{bmatrix} -\frac{i+1}{2} & \frac{i-1}{2} & 1 & 0 \end{bmatrix}^t \right\}$.

OR

Here $\overline{A} = \begin{bmatrix} 1 & -1 & 1 & 0 \\ -i & -1 & 0 & 1 \end{bmatrix}$. Then $\mathbf{v} = [a, b, c, d]^t \in \text{null}(\overline{A})$ iff $a + b + c = 0$ and $-ia - b + d = 0$. Note that c and d can be considered as leading variables (how?). These gives $c = -a - b$ and $d = ia + b$. Thus

$$\mathbf{v} = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} a \\ b \\ -a-b \\ ia+b \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \\ -1 \\ i \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}.$$

From this, a basis for $V^\perp = \text{null}(\overline{A})$ is obtained as $\{[1 \ 0 \ -1 \ i]^t, [0 \ 1 \ -1 \ 1]^t\}$.

□

50. Let A be a 2×2 orthogonal matrix. Show that there exists a real number θ such that $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ or $A = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}$. In the first case, A rotates the vectors of $\mathbb{R}^2$ by the angle θ counterclockwise, and in the second case, A reflects the vectors of $\mathbb{R}^2$ about a line; in this case find the line.

Solution: As columns of A have norm 1, we may assume that they are $[\cos \theta, \sin \theta]^t$ and $[\cos \phi, \sin \phi]^t$. As columns are orthogonal, we get $\cos(\phi - \theta) = 0 \Rightarrow \phi = \theta + \frac{\pi}{2}$ or $\phi = \theta + \frac{3\pi}{2}$.

Substituting the value of ϕ , we find that $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ or $A = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}$.

In the first case $A[r \cos \alpha, r \sin \alpha]^t = [r \cos(\theta + \alpha), r \sin(\theta + \alpha)]^t$. Therefore it is the rotation by the angle θ counterclockwise.

In the second case $A[r \cos \alpha, r \sin \alpha]^t = [r \cos(\theta - \alpha), r \sin(\theta - \alpha)]^t$, and therefore it is the reflection about the line $y = \tan\left(\frac{\theta}{2}\right)x$. □

51. Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be an orthonormal basis of $\mathbb{C}^n$. Show that for any $\mathbf{x}, \mathbf{y} \in \mathbb{C}^n$,

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^n (\mathbf{x} \cdot \mathbf{v}_i)(\mathbf{v}_i \cdot \mathbf{y}) \quad \text{and} \quad \mathbf{x} \cdot \mathbf{x} = \sum_{i=1}^k |(\mathbf{x} \cdot \mathbf{v}_i)|^2 \quad \text{for } 1 \leq k \leq n.$$

Further, show that $\mathbf{x} \cdot \mathbf{x} = \sum_{i=1}^k |(\mathbf{x} \cdot \mathbf{v}_i)|^2$ iff $\mathbf{x} \in \text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$, where $1 \leq k \leq n$.

Solution: We have $\mathbf{x} = \sum_{i=1}^n \alpha_i \mathbf{v}_i$ and $\mathbf{y} = \sum_{i=1}^n \beta_i \mathbf{v}_i$, where $\alpha_i = \mathbf{x} \cdot \mathbf{v}_i, \beta_i = \mathbf{v}_i \cdot \mathbf{y}$ for $i = 1, 2, \dots, n$.

Therefore

$$\mathbf{x} \cdot \mathbf{y} = \left(\sum_{i=1}^n \alpha_i \mathbf{v}_i \right) \cdot \left(\sum_{i=1}^n \beta_i \mathbf{v}_i \right) = \sum_{i,j=1}^n \alpha_i \beta_j (\mathbf{v}_i \cdot \mathbf{v}_j) = \sum_{i=1}^n \alpha_i \beta_i = \sum_{i=1}^n (\mathbf{x} \cdot \mathbf{v}_i)(\mathbf{v}_i \cdot \mathbf{y}).$$

Also

$$\mathbf{x} \cdot \mathbf{x} = \sum_{i=1}^n (\mathbf{x} \cdot \mathbf{v}_i)(\mathbf{v}_i \cdot \mathbf{x}) = \sum_{i=1}^n |(\mathbf{x} \cdot \mathbf{v}_i)|^2 \geq \sum_{i=1}^k |(\mathbf{x} \cdot \mathbf{v}_i)|^2 \text{ for } 1 \leq k \leq n.$$

Now, $\mathbf{x} \cdot \mathbf{x} = \sum_{i=1}^k |(\mathbf{x} \cdot \mathbf{v}_i)|^2 \Rightarrow \sum_{i=k+1}^n |(\mathbf{x} \cdot \mathbf{v}_i)|^2 = 0 \Rightarrow \alpha_i = \mathbf{v}_i \cdot \mathbf{x} = 0$ for $i = k+1, \dots, n$. Therefore

$$\mathbf{x} = \sum_{i=1}^k (\mathbf{v}_i \cdot \mathbf{x}) \mathbf{v}_i \in \text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k).$$

Conversely, $\mathbf{x} \in \text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) \Rightarrow \mathbf{x} = \sum_{i=1}^k (\mathbf{v}_i \cdot \mathbf{x}) \mathbf{v}_i$, and therefore $\mathbf{x} \cdot \mathbf{x} = \sum_{i=1}^k |(\mathbf{x} \cdot \mathbf{v}_i)|^2$. $\square$

52. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation and let $A = [a_{ij}]$ be the matrix of T with respect to an orthonormal basis $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ of $\mathbb{R}^n$. Show that $a_{ij} = \mathbf{v}_i \cdot (T\mathbf{v}_j)$ for all i, j .

Solution: For $1 \leq j \leq n$, we have

$$\begin{aligned} T\mathbf{v}_j &= a_{1j}\mathbf{v}_1 + a_{2j}\mathbf{v}_2 + \dots + a_{ij}\mathbf{v}_i + \dots + a_{nj}\mathbf{v}_n \\ \Rightarrow \mathbf{v}_i \cdot (T\mathbf{v}_j) &= \mathbf{v}_i \cdot (a_{1j}\mathbf{v}_1 + a_{2j}\mathbf{v}_2 + \dots + a_{ij}\mathbf{v}_i + \dots + a_{nj}\mathbf{v}_n), \quad (1 \leq i \leq n) \\ \Rightarrow \mathbf{v}_i \cdot (T\mathbf{v}_j) &= a_{1j}(\mathbf{v}_i \cdot \mathbf{v}_1) + a_{2j}(\mathbf{v}_i \cdot \mathbf{v}_2) + \dots + a_{ij}(\mathbf{v}_i \cdot \mathbf{v}_i) + \dots + a_{nj}(\mathbf{v}_i \cdot \mathbf{v}_n) \\ \Rightarrow \mathbf{v}_i \cdot (T\mathbf{v}_j) &= a_{ij}. \end{aligned}$$

$\square$

53. Let W be a subspace of $\mathbb{C}^n$ and let $\mathbf{u} \in \mathbb{C}^n$. Show that $\mathbf{v} \in W$ is the projection of $\mathbf{u}$ onto W if and only if $\|\mathbf{u} - \mathbf{v}\| \leq \|\mathbf{u} - \mathbf{w}\|$ for every $\mathbf{w} \in W$.

Solution: Let $\mathbf{v} \in W$ and $\mathbf{u} = \mathbf{v} + \mathbf{v}'$, where $\mathbf{v}' \in W^\perp$. Then $\mathbf{u} - \mathbf{v} \in W^\perp$. Thus, $\mathbf{v} - \mathbf{w} \in W$ for every $\mathbf{w} \in W$, and hence $(\mathbf{u} - \mathbf{v}) \cdot (\mathbf{v} - \mathbf{w}) = 0$. Therefore,

$$\begin{aligned} \|\mathbf{u} - \mathbf{w}\|^2 &= \|(\mathbf{u} - \mathbf{v}) + (\mathbf{v} - \mathbf{w})\|^2 = \|\mathbf{u} - \mathbf{v}\|^2 + \|\mathbf{v} - \mathbf{w}\|^2 \\ \Rightarrow \|\mathbf{u} - \mathbf{v}\| &\leq \|\mathbf{u} - \mathbf{w}\|. \end{aligned}$$

For the converse, suppose, if possible, $\mathbf{v}_1$ is the projection of $\mathbf{u}$ onto W and $\mathbf{v}_1 \neq \mathbf{v}$. Then we have $(\mathbf{u} - \mathbf{v}_1) \cdot (\mathbf{v}_1 - \mathbf{v}) = 0$ and therefore

$$\begin{aligned} \|\mathbf{u} - \mathbf{v}\|^2 &= \|(\mathbf{u} - \mathbf{v}_1) + (\mathbf{v}_1 - \mathbf{v})\|^2 \\ &= \|\mathbf{u} - \mathbf{v}_1\|^2 + \|\mathbf{v}_1 - \mathbf{v}\|^2 \\ &> \|\mathbf{u} - \mathbf{v}_1\|^2, \text{ a contradiction.} \end{aligned}$$

Hence $\mathbf{v}$ is the projection of $\mathbf{u}$ onto W . $\square$

54. Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ be an orthogonal set of non-zero vectors in $\mathbb{C}^n$. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_q$ be some vectors in $\mathbb{C}^n$ that are orthogonal to S . If $p + q > n$ then show that the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_q$ are linearly dependent.

Solution: Since $p + \dim S^\perp = \dim V = n$, we have $\dim S^\perp = n - p < q$. Thus, any set of more than $n - p$ vectors in $S^\perp$ are linearly dependent. Note that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_q\} \subseteq S^\perp$, and hence they must be linearly dependent. $\square$

55. Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n, n \geq 2$ such that $\|\mathbf{v}\| = \|\mathbf{w}\| = 1$. Prove that there exists an orthogonal matrix A such that $A(\mathbf{v}) = \mathbf{w}$ and $\det(A) = 1$.

Solution: Consider a basis for $\mathbb{R}^n$ containing $\mathbf{w}$ and apply Gram-Schmidt Process to obtain an orthonormal basis $\{\mathbf{w}, \mathbf{w}_2, \mathbf{w}_3, \dots, \mathbf{w}_n\}$. Let $B = [\mathbf{w} \ \mathbf{w}_2 \ \mathbf{w}_3 \ \dots \ \mathbf{w}_n]$. Then B is an orthogonal matrix and $B\mathbf{e}_1 = \mathbf{w}$. Similarly, we find an orthogonal matrix C such that $C\mathbf{e}_1 = \mathbf{v}$. Now take $A = BC^{-1}$, which is an orthogonal matrix and satisfies $A\mathbf{v} = \mathbf{w}$.

Since A is an orthogonal matrix, we must have $\det(A) = \pm 1$. If $\det(A) = 1$, then we are done. If $\det(A) = -1$, then consider the matrix $\tilde{C}$ obtained on multiplying the last column of C by -1 . Now take $\tilde{A} = B\tilde{C}^{-1}$. We find that $\tilde{A}\mathbf{v} = \mathbf{w}$ and $\det(\tilde{A}) = 1$. $\square$

56. (Spectral Decomposition Theorem) Let A be a normal matrix. Show that there are distinct complex numbers $\alpha_1, \dots, \alpha_k$ and matrices $E_1, \dots, E_k$ such that $A = \alpha_1 E_1 + \dots + \alpha_k E_k$, where $E_1 + \dots + E_k = I$, $E_i E_j = O$ for $i \neq j$ and $E_i^2 = E_i$ for $i = 1, \dots, k$.

Solution: Let the distinct eigenvalues of A be $\alpha_1, \dots, \alpha_k$. Let an orthonormal basis of eigenvectors of A corresponding to eigenvalue α_i be $\{\mathbf{u}_{i1}, \dots, \mathbf{u}_{ir_i}\}$. Now take $E_i = \mathbf{u}_{i1}\mathbf{u}_{i1}^* + \dots + \mathbf{u}_{ir_i}\mathbf{u}_{ir_i}^*$.

Let D be the diagonal matrix whose first r_1 diagonal entries are α_1 , then next r_2 diagonal entries are α_2 , and so on, that is, $D = \text{diag}[\alpha_1, \dots, \alpha_1, \alpha_2, \dots, \alpha_2, \dots, \alpha_k, \dots, \alpha_k]$. Let $Q = [\mathbf{u}_{11} \ \dots \ \mathbf{u}_{1r_1} \ \dots \ \dots \ \mathbf{u}_{k1} \ \dots \ \mathbf{u}_{kr_k}]$, so that Q is unitary. Then we have

$$\begin{aligned} A &= QDQ^* = [\alpha_1 \mathbf{u}_{11} \ \dots \ \alpha_1 \mathbf{u}_{1r_1} \ \dots \ \dots \ \alpha_k \mathbf{u}_{k1} \ \dots \ \alpha_k \mathbf{u}_{kr_k}] Q^* \\ &= \alpha_1 (\mathbf{u}_{11}\mathbf{u}_{11}^* + \dots + \mathbf{u}_{1r_1}\mathbf{u}_{1r_1}^*) + \dots + \alpha_k (\mathbf{u}_{k1}\mathbf{u}_{k1}^* + \dots + \mathbf{u}_{kr_k}\mathbf{u}_{kr_k}^*) \\ &= \alpha_1 E_1 + \dots + \alpha_k E_k. \end{aligned}$$

Further,

$$E_1 + \dots + E_k = (\mathbf{u}_{11}\mathbf{u}_{11}^* + \dots + \mathbf{u}_{1r_1}\mathbf{u}_{1r_1}^*) + \dots + (\mathbf{u}_{k1}\mathbf{u}_{k1}^* + \dots + \mathbf{u}_{kr_k}\mathbf{u}_{kr_k}^*) = QQ^* = I.$$

For $i \neq j$, the vector $\mathbf{u}_{is}$ is orthogonal to the vector $\mathbf{u}_{jt}$ for each $s = 1, \dots, r_i$ and $t = 1, \dots, r_j$, that is, $\mathbf{u}_{is}^* \mathbf{u}_{jt} = 0$. Therefore

$$E_i E_j = (\mathbf{u}_{i1}\mathbf{u}_{i1}^* + \dots + \mathbf{u}_{ir_i}\mathbf{u}_{ir_i}^*)(\mathbf{u}_{j1}\mathbf{u}_{j1}^* + \dots + \mathbf{u}_{jr_j}\mathbf{u}_{jr_j}^*) = \sum_{s=1}^{r_i} \sum_{t=1}^{r_j} \mathbf{u}_{is}\mathbf{u}_{is}^* \mathbf{u}_{jt}\mathbf{u}_{jt}^* = 0.$$

Finally, as $\{\mathbf{u}_{i1}, \dots, \mathbf{u}_{ir_i}\}$ is an orthonormal set, we have

$$\begin{aligned} E_i^2 &= \sum_{s=1}^{r_i} \sum_{t=1}^{r_i} \mathbf{u}_{is}\mathbf{u}_{is}^* \mathbf{u}_{it}\mathbf{u}_{it}^* = \sum_{s=1}^{r_i} \mathbf{u}_{is}\mathbf{u}_{is}^* \mathbf{u}_{is}\mathbf{u}_{is}^*, \text{ as } \mathbf{u}_{is}^* \mathbf{u}_{it} = 0 \text{ for } s \neq t \\ &= \sum_{s=1}^{r_i} \mathbf{u}_{is}\mathbf{u}_{is}^*, \text{ as } \mathbf{u}_{is}^* \mathbf{u}_{is} = 1 \\ &= E_i. \end{aligned}$$

$\square$

57. Orthogonally diagonalize, and then find spectral decomposition of the matrix $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$.

Solution: Let A denote the given matrix. Solving the equation $\det(A - \lambda I) = 0$, we get $\lambda = 4, 1, 1$. So the eigenvalues of A are 4, 1 and 1.

To find the eigenvectors of A corresponding to the eigenvalue 4:

We have

$$\begin{aligned}
 A - 4I = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} &\xrightarrow{R_2 \rightarrow 2R_2 + R_1} \begin{bmatrix} -2 & 1 & 1 \\ 0 & -3 & 3 \\ 1 & 1 & -2 \end{bmatrix} \\
 &\xrightarrow{R_3 \rightarrow 2R_3 + R_1} \begin{bmatrix} -2 & 1 & 1 \\ 0 & -3 & 3 \\ 0 & 3 & -3 \end{bmatrix} \\
 &\xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{bmatrix} -2 & 1 & 1 \\ 0 & -3 & 3 \\ 0 & 0 & 0 \end{bmatrix} \\
 &\xrightarrow{R_2 \rightarrow \frac{1}{3}R_2} \begin{bmatrix} -2 & 1 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.
 \end{aligned}$$

Thus, if $\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, then the system $A\mathbf{x} = 4\mathbf{x}$ is equivalent to $2x = y + z$ and $y = z$. Letting

$y = z = 1$, we get $x = 1$. So $\{\mathbf{q}_1\}$, where $\mathbf{q}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, forms a basis of the solution space

of $A\mathbf{x} = 4\mathbf{x}$. Therefore $\{\mathbf{u}_1\}$, where $\mathbf{u}_1 = \frac{\mathbf{q}_1}{\|\mathbf{q}_1\|} = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$, is an orthonormal basis of the eigenspace of A corresponding to the eigenvalue 4.

To find the eigenvectors of A corresponding to the eigenvalue 1:

We have

$$\begin{aligned}
 A - I = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} &\xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \\
 &\xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.
 \end{aligned}$$

Thus, if $\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, then the system $A\mathbf{x} = 1\mathbf{x}$ is equivalent to $x + y + z = 0$. Letting

$y = 0, z = 1$, we get $x = -1$. Letting $y = 1, z = 0$, we get $x = -1$. So $\{\mathbf{q}_2, \mathbf{q}_3\}$, where $\mathbf{q}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ and $\mathbf{q}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$, form a basis of the solution space of $A\mathbf{x} = 1\mathbf{x}$.

However, $\{\mathbf{q}_2, \mathbf{q}_3\}$ is not an orthogonal set. Using Gram-Schmidt Process, we write

$$\mathbf{u}_2 = \frac{\mathbf{q}_2}{\|\mathbf{q}_2\|} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} \text{ and } \mathbf{u}_3 = \frac{\mathbf{q}_3 - (\mathbf{u}_2 \cdot \mathbf{q}_3)\mathbf{u}_2}{\|\mathbf{q}_3 - (\mathbf{u}_2 \cdot \mathbf{q}_3)\mathbf{u}_2\|} = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ -1/\sqrt{6} \end{bmatrix}.$$

Therefore $\{\mathbf{u}_2, \mathbf{u}_3\}$ is an orthonormal basis of the eigenspace of A corresponding to the eigenvalue 1.

The set $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ of eigenvectors of A is clearly orthonormal. Thus we have $Q^t A Q = D$,

$$\text{where } Q = [\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3] = \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \end{bmatrix} \text{ and } D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Here $\lambda_1 = 4, \lambda_2 = 1 = \lambda_3$ and $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ are the orthonormal eigenvectors corresponding to the eigenvalues 4, 1 and 1. So, the required spectral decomposition of A is

$$A = 4.\mathbf{u}_1\mathbf{u}_1^t + 1.(\mathbf{u}_2\mathbf{u}_2^t + \mathbf{u}_3\mathbf{u}_3^t) = 4. \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} + 1. \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{bmatrix}.$$

OR

We can **also** get two orthogonal eigenvectors of A corresponding to the eigenvalue 1 as follows.

The vector $\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is an eigenvector of A corresponding to the eigenvalue 1 iff $x + y + z = 0$.

One such eigenvalue is $\mathbf{q}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$. If $\mathbf{q}_3 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ is another eigenvector of A corresponding to the eigenvalue 1 such that $\mathbf{q}_3 \cdot \mathbf{q}_2 = 0$, then we have $a + b + c = 0$ and $-a + c = 0$. This gives $c = a$ and $b = -2a$. One such vector is $\mathbf{q}_3 = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$.

□

58. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of a matrix A . Show that $\det(A) = \prod_{i=1}^n \lambda_i$ and $\text{trace}(A) = \sum_{i=1}^n \lambda_i$.

Solution: By Schur's Theorem, the matrix A is unitarily similar to an upper triangular matrix T , and that the diagonal entries of T are the eigenvalues of A . Let Q be a unitary matrix such that $A = Q^* T Q$ and $\lambda_1, \dots, \lambda_n$ be the diagonal entries of T . It is clear that $\det(T) = \prod_{i=1}^n \lambda_i$ and $\text{trace}(T) = \sum_{i=1}^n \lambda_i$. Using the facts $\det(XY) = \det(YX)$, $\text{trace}(XY) = \text{trace}(YX)$ and $QQ^* = I$, we have

$$\det(A) = \det(Q^* T Q) = \det(Q Q^* T) = \det(T) = \prod_{i=1}^n \lambda_i$$

and

$$\text{trace}(A) = \text{trace}(Q^* T Q) = \text{trace}(Q Q^* T) = \text{trace}(T) = \sum_{i=1}^n \lambda_i.$$

□

59. Find a 3×3 symmetric matrix with eigenvalues 1, 2 and 3, and corresponding eigenvectors

$$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \text{ and } \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}.$$

Solution: Observe that the given three vectors are pairwise orthogonal. Multiplying these vectors by the reciprocals of their corresponding norm, we get the orthonormal eigenvectors

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{u}_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \text{ and } \mathbf{u}_3 = \frac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}. \text{ Therefore, the required matrix is}$$

$$\begin{aligned} A = 1.\mathbf{u}_1\mathbf{u}_1^t + 2.\mathbf{u}_2\mathbf{u}_2^t + 3.\mathbf{u}_3\mathbf{u}_3^t &= \frac{1}{2} \cdot \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \frac{2}{3} \cdot \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} + \frac{1}{2} \cdot \begin{bmatrix} 1 & -1 & -2 \\ -1 & 1 & 2 \\ -2 & 2 & 4 \end{bmatrix} \\ &= \begin{bmatrix} \frac{5}{3} & -\frac{2}{3} & -\frac{1}{3} \\ -\frac{2}{3} & \frac{5}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3} & \frac{8}{3} \end{bmatrix}. \end{aligned}$$

□

60. Let $\mathbf{q}$ be a unit vector in $\mathbb{R}^n$ and $W = \text{span}(\mathbf{q})$. Show that the orthogonal projection of a vector $\mathbf{v}$ onto W is given by $(\mathbf{q}\mathbf{q}^t)\mathbf{v}$, and that the matrix of this projection map is $\mathbf{q}\mathbf{q}^t$.

Solution: Note that $\{\mathbf{q}\}$ is a basis of W and $\|\mathbf{q}\| = 1$. By definition, $\text{proj}_W(\mathbf{v}) = (\mathbf{q} \cdot \mathbf{v}) \frac{\mathbf{q}}{\|\mathbf{q}\|^2} = (\mathbf{q}^t \mathbf{v}) \mathbf{q}$. As $\mathbf{q}^t \mathbf{v}$ is a scalar, it commutes with any matrix, and so, $(\mathbf{q}^t \mathbf{v}) \mathbf{q} = \mathbf{q}(\mathbf{q}^t \mathbf{v}) = (\mathbf{q}\mathbf{q}^t)\mathbf{v}$.

Thus the orthogonal projection of a vector $\mathbf{v}$ onto W is given by a linear map $P : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by $P(\mathbf{v}) = (\mathbf{q}\mathbf{q}^t)\mathbf{v}$. [Show that P is indeed linear!]. Recall that with respect to the standard basis, the matrix of T defined by $T(\mathbf{v}) = A\mathbf{v}$ is A . Hence the matrix of P is $\mathbf{q}\mathbf{q}^t$. □